When any cells in a contiguous set of 5 cells are being balanced, those affected cells are measured in a reduced 12.5-ms decimation period, to allow the cell balancing to function properly without affecting the integrated OV and UV protections. Since cell balancing is typically only performed during pack charge or idle periods, the shortened decimation periods should not impact accuracy as the system noise during these times is greatly reduced. This reduced decimation period is only applied to sets where one of the cells is being balanced. The following summarizes this for the BQ76920 – BQ76940 devices:

- VC1 to VC5 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL1 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL1 register are 1.
- VC6 to VC10 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL2 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL2 register are 1.
- VC11 to VC15 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL3 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL3 register are 1.
- Total update interval is 250 ms.

Each differential cell input is factory-trimmed for gain or offset, such that the resulting reading through I^2C is always consistent from part-to-part and requires no additional calibration or correction factor application.

The ADC is required to be enabled in order for the integrated OV and UV protections to be operating.

The following shows how to convert the 14-bit ADC reading into an analog voltage. Each device is factory calibrated, with a GAIN and OFFSET stored into EEPROM.

The ADC transfer function is a linear equation defined as follows:

$$\mathbf{V(cell) = GAIN \times ADC(cell) + OFFSET} \tag{1}$$

GAIN is stored in units of µV/LSB, while OFFSET is stored in mV units.

Some example cell voltage calculations are provided in the table below. For illustration purposes, the example uses a hypothetical GAIN of 380 µV/LSB (ADCGAIN<4:0> = 0x0F) and OFFSET of 30 mV (ADCOFFSET<7:0> = 0x1E).

14-Bit ADC Result	ADC Result in Decimal	GAIN (µV/LSB)	OFFSET (mV)	Cell Voltage (mV)
0x1800	6144	380	30	2365
0x1F10	7952	380	30	3052

备注

When entering NORMAL mode from SHIP mode, please allow for the following times before reading out initial cell voltage data:

BQ76920: 250 ms

BQ76930: 400 ms

BQ76940: 800 ms